

Introduction

Insertion Sort is a fundamental concept in Data Structures and Algorithms used for solving computational problems efficiently.

Definition

Insertion Sort refers to the method or structure used to organize, process, and manage data effectively.

Working Principle

The working of Insertion Sort involves step-by-step execution following a defined logic or pattern depending on the input.

Algorithm / Steps

1. Start 2. Process input 3. Apply logic 4. Return output 5. End

Example

A simple example of Insertion Sort helps in understanding its behavior with real inputs.

Advantages

- Easy to understand
- Efficient in many scenarios
- Widely used in applications

Disadvantages

- May be slow for large inputs
- Performance varies based on input

Time Complexity

- Best Case: $O(n)$
- Average Case: $O(n \log n)$
- Worst Case: $O(n^2)$

Space Complexity

Depends on implementation, generally $O(1)$ or $O(n)$

Applications

- Searching
- Sorting
- Data processing
- Real-world systems

Interview Questions

- Explain Insertion Sort
- What is its complexity?
- When to use it?

Conclusion

Insertion Sort is essential for mastering problem solving and technical interviews.

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